



Security index based on perfectly undetectable attacks: Graph-theoretic conditions[☆]

Sebin Gracy^{*}, Jezdimir Milošević, Henrik Sandberg

The Division of Decision and Control Systems, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden

ARTICLE INFO

Article history:

Received 13 March 2020

Received in revised form 31 May 2021

Accepted 5 August 2021

Available online 1 October 2021

ABSTRACT

The notion of security index quantifies the *least* effort involved in conducting perfectly undetectable attacks. Thus, the security index enables a systems operator to assess the vulnerability of a component, informs sensor placement strategies, and helps in deciding the feasibility of secure estimators and fault detectors. In this paper, we investigate the (possible) variation in this index as a consequence of variation in the system parameters. To this end, we adopt a structured systems approach, typically represented by a directed graph, with the edges of the said graph being in one-to-one correspondence with the system parameters. We first show that the security index is generic. That is, for almost all choices of edge weights, the security index of a component remains the same. We refer to such an index as the generic security index. Secondly, we derive graph-theoretic conditions (and based on those an algorithm) for computing the generic security index. Third, we provide graph-theoretic conditions for computing lower (resp. upper) bounds on the values that the security index of a component can take for all nonzero choices of the edge weights of the directed graph. Finally, we provide a brute force search method for calculating the said bounds.

© 2021 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Cyber-Physical Systems (CPS) has become ubiquitous in the present times; including, but not limited to, smart grid (Amin & Wollenberg, 2005), sensor networks (Sinopoli, Sharp, Schenato, Schaffert, & Sastry, 2003) and formation control of autonomous vehicles (Olfati-Saber & Murray, 2002). Given such widespread prevalence of CPS, it is fair to say that the smooth functioning of CPS is sine qua non for the modern society. Indeed, any significant malfunctioning – deliberate (possibly by injecting malicious inputs into the system) or otherwise (for instance faulty actuators, sensors etc.) – of CPS has severe consequences, as evidenced by, to cite a few, the Stuxnet attack (Kushner, 2013), blackout in northeast US (Andersson et al., 2005), and the Maroochy water incident (Slay & Miller, 2007).

[☆] This work was supported in part by the Swedish Civil Contingencies Agency (project CERES), and the Swedish Research Council (project 2016-00861). The material in this paper was partially presented at the 2020 American Control Conference, July 1–3, 2020, Denver, CO, USA. This paper was recommended for publication in revised form by Associate Editor Shun-ichi Azuma under the direction of Editor Thomas Parisini.

^{*} Corresponding author.

E-mail addresses: gracy@kth.se (S. Gracy), jezdimir@kth.se (J. Milošević), hsan@kth.se (H. Sandberg).

The security of CPS has attracted the attention of researchers across different areas such as automatic control, computer security, and sensor network security. The control systems community has made significant progress on security of CPS in the last decade; see for instance (Giraldo, Sarkar, Cárdenas, Maniatakos, & Kantarcioglu, 2017; Mo et al., 2012). In particular, problems which were previously considered in an attack-free setting, have now been extended to account for cyber-attacks; for instance design of attack resilient estimators (Fawzi, Tabuada, & Diggavi, 2014), detectors (Pasqualetti, Bicchi, & Bullo, 2012), and consensus protocols (Sundaram & Hadjicostis, 2010). Since resources to secure CPS against malicious attacks are often limited, it becomes imperative to *identify, assess and minimize* the risk of threats to their proper functioning. The so-called risk management framework is a technique for achieving the aforementioned objectives; see Chong, Sandberg, and Teixeira (2019) for a detailed overview. In this context, Deghat, Ugrinovskii, Shames, and Langbort (2019), Mo, Garone, Casavola, and Sinopoli (2010), Pasqualetti et al. (2012) identify non-trivial attack scenarios in deterministic settings, while Bai, Pasqualetti, and Gupta (2017) does the same in stochastic settings. Building off of the work that identifies (non-trivial) attack scenarios, quantitative tools, like security index, for likelihood assessment of attacks have been provided in Milošević, Teixeira, Johansson, and Sandberg (2020), Sandberg and Teixeira (2016), Tang, Kuijper, Chong, Mareels, and Leckie (2019). Finally, the quantification of the worst possible

damage that such attacks can cause to the system has been provided in, for instance, Ahmed, Murguía, and Ruths (2017), Bai, Pasqualetti, and Gupta (2015), Murguía, Shames, Ruths, and Nesic (2018). The present paper is focused on the notion of security index in dynamical systems.

For a given component i ,¹ the security index is equal to the minimum number of sensors and actuators that needs to be compromised in order to execute a given type of attack targeting i . A large (resp. small) security index means that a large (resp. small) number of components have to be attacked in conjunction with the component of interest. It is thus immediate that the notion of security index serves as a proxy for the least effort involved in conducting a severe attack using a given component. Consequently, this notion can be leveraged in several contexts as follows. First, the security index helps the system operator to identify *which* of the components are the most vulnerable in the system. Armed with this information, (s)he can then accordingly prioritize allocation of security investment (Vuković, Sou, Dan, & Sandberg, 2012). Furthermore, the system operator has to secure not only the *critical set* of measurements,² but (possibly) also those components that have a low security index. Second, since components with a low security index are, by definition, more vulnerable to malicious attacks, sensor placement strategies have been proposed in the literature for increasing the aforementioned index (Milošević et al., 2020, Section V.C). Third, the notion of security index also has implications for the possibility of the operator to reconstruct a detectable attack, which in turn is useful for a deciding feasibility of secure estimators and fault detectors (Sandberg & Teixeira, 2016).

The notion of security index for dynamical linear network systems has been studied since (Sandberg & Teixeira, 2016). Note that the definition of security index in Sandberg and Teixeira (2016) relies on the concept of *undetectable attacks*. Undetectable attacks are those where the output(s) are manipulated so as to correspond to output(s) generated as a consequence of natural disturbance (Pasqualetti et al., 2012). A more dangerous class of attacks is that of *perfectly undetectable attacks* (Weerakkody, Liu, Son, & Sinopoli, 2016), i.e., attacks which do not leave a trace at the output. Clearly, perfectly undetectable attacks could have severe ramifications. Hence, from a security standpoint, it is more prudent to define the notion of security index with respect to perfectly undetectable attacks. Towards this end, for a given component i , a novel security index has been developed in Milošević et al. (2020). Notably, in Milošević et al. (2020), the authors consider attacks on *both* actuators and sensors. Practical motivations for attacks on both actuators and sensors have been provided in Mohsenian-Rad and Leon-Garcia (2011).

Observe that Milošević et al. (2020) defines the security index with respect to a given plant model, i.e., each of the system matrices having specific numerical entries. However, for modern infrastructure networks, as a consequence of their complexity, the coefficients of the system matrices need not necessarily remain fixed throughout. Note that changing the coefficients in the system matrices might (or might not) cause the value of the security index of a component to change. Hence, we would like to characterize the vulnerability of a component not by relying on the numerical entries of the system matrices, but rather by solely leveraging the knowledge of the topology of the underlying graph. More precisely, we aim to i) develop a notion of security index that is impervious to variations in system parameters, and

(ii) provide bounds on the values that the security index of a component can take for all nonzero choices of edge weights³ of the underlying graph.

We consider a structured systems framework. That is, the state, input, output and feedthrough matrices are *structured* matrices; certain positions are fixed to zero, while the positions that are not a priori fixed to zero are referred to as *free parameters*. Each numerical choice of free parameters yields a *realization* of the system at hand. Against such a backdrop, our contributions are as follows: First, we show that for a given component i , the security index is *generic*. That is, the security index of a component i has the same value for *almost all* choices of free parameters. Almost all choices of parameters mean all choices of the parameter except those lying in some proper algebraic variety of the space of free parameters. We refer to this value as the generic security index of a component i . Second, we provide graph-theoretic conditions, both in terms of linkings on the directed graph, and matchings on appropriately-defined bipartite graph, for computing the generic security index, and, based on the latter, an algorithm for the same. Notice that the generic security index provides only *probabilistic* guarantees of security, whereas there maybe scenarios in which an attack can cause deep impacts. Hence, there is also a need to obtain *deterministic* guarantees of security. Towards this end – as our third contribution – we establish graph-theoretical conditions that aid in computing lower (resp. upper) bounds on the value that the security index of a component can take for *all* nonzero choices of edge weights of the underlying graph. The said conditions are in terms of uniquely restricted matchings on appropriately-defined bipartite graphs. Moreover, we show that computing either of the aforementioned bounds is NP-hard.

Recently, the notion of *robust security index* of a component has been proposed and developed, based on structural representation of the given system (Milošević et al., 2020). The work in Milošević et al. (2020) differs with the present paper in the following sense. First, the notion of robust security index is developed for finding only the most vulnerable actuators in the system, that is, those actuators that are vulnerable for *every* admissible realization of the system parameters. Yet, some actuators can be vulnerable for almost all, but not necessarily for *every* realization of the system parameters. Hence, we aim to develop an index that is less strict than the robust security index, yet still robust to numerical changes in modeling parameters – the notion of generic security index addresses this gap. Second, the (exact) computation of the robust security index in Milošević et al. (2020) relies on the assumption that the input matrix has a particular structure, whereas the upper (resp. lower) bound on the security index for all admissible realizations that we derive accounts for arbitrary structures of input matrices. Third, the lower bound on the security index for all admissible realizations that we establish is also a lower bound for the robust security index in Milošević et al. (2020).

A preliminary version of this paper appeared in Gracy, Milošević, and Sandberg (2020). The present paper differs from Gracy et al. (2020) in the following ways: (i) we provide, in terms of matchings on appropriately defined bipartite graphs, graph-theoretic conditions for computing the generic security index, and (ii) we provide graph-theoretic conditions, in terms of existence of uniquely restricted matchings of appropriate size, for computing upper (resp. lower) bounds on the security index of a component for all nonzero choices of edge weights of the underlying graph. The said bounds provide deterministic guarantees of security

¹ A component can be a sensor or an actuator.

² Critical measurements are those which if lost/corrupted would render the network unobservable. That is, given the output measurements, the states can no longer be uniquely determined; see Sou, Sandberg, and Johansson (2012) for more details.

³ A choice of edge weights is said to be *nonzero* if every edge has a nonzero weight.

Paper outline:

The paper is organized as follows: In the remainder of this section, we list the various notations used in the sequel. The problem being investigated is made precise in Section 2. The background material needed for developing the main results are gathered in Section 3. The main results, together with their relevant interpretation, are provided in Section 4. The rigorous analysis of the results in Section 4 has been deferred to the Appendix. Illustration of the theoretical findings of the present paper on a practical example, and an example that exhibits the effectiveness of our algorithm for computing generic security index (Algorithm 1) on a realistic setting in the context of risk assessment, are given in Gracy, Milošević, and Sandberg (2021). Finally, we summarize the paper in Section 5.

Notations

Let $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}$, $\mathbb{Z}_+$ and $\mathbb{Z}_{\geq 0}$ denote the set of real numbers, complex numbers, integers, positive integers, and non-negative integers, respectively. I_N indicates an identity matrix of size N . $|\mathcal{X}|$ indicates cardinality of a set $\mathcal{X}$. For a column vector a , $a^{(i)}$ denotes the element corresponding to its i th row.

2. Problem formulation

In this section, we first present the system under consideration, recall the formulation of the security index δ and motivate the need for structured systems framework, explain how a directed graph can be associated with a structured system, and finally present the problem(s) of interest.

2.1. System model

The plant dynamics under attack are modeled by

$$\begin{aligned} x(k+1) &= Wx(k) + Bu(k) + B_a a(k), \\ y(k) &= Cx(k) + D_a a(k). \end{aligned} \quad (1)$$

Here, $x(k) \in \mathbb{R}^N$ are the plant states, $u(k) \in \mathbb{R}^Q$ are the control inputs, $a(k) \in \mathbb{R}^P$ are the attacks and $y(k) \in \mathbb{R}^M$ are the sensor measurements. We denote by $\mathcal{S}$ the set of protected sensors,⁴ and by $\mathcal{I}$ the joint set of actuators and unprotected sensors, and attack cardinality by

$$\|a\|_0 = |\cup_{k \in \mathbb{Z}_{\geq 0}} \text{supp}(a(k))|,$$

where $\text{supp}(a(k)) = \{i \in \mathcal{I} : a^{(i)}(k) \neq 0\}$. The vector $a(k)$ models both actuator and sensor attacks. Particularly, the first $Q < P$ elements of $a(k)$ correspond to attacks on the actuator(s), whereas the remaining elements correspond to attacks on the sensor(s). Thus, we have

$$B_a = \begin{bmatrix} B & 0 \end{bmatrix}, \quad D_a = \begin{bmatrix} 0 & I_{M-|\mathcal{S}|} \\ 0 & 0 \end{bmatrix}.$$

Note that, in $y(k)$, the last $|\mathcal{S}|$ measurements correspond to those of the protected sensors, while the first $M - |\mathcal{S}|$ measurements correspond to those of the unprotected sensors.

Our focus is on *perfectly undetectable attacks* (Weerakkody et al., 2016). These attacks are extremely dangerous, because the attacker does not leave traces in the measurements. In that way, the attacker can endanger system components or make some of the system states diverge from the steady state by an arbitrary value without the operator(s) noticing.

⁴ Protected sensors are those, as the name implies, that cannot be manipulated by an attacker. The protection is typically accomplished by implementing encryption schemes and/or improving physical protection (Vuković et al., 2012).

Definition 1. Let $y(x(0), u, a)$ denote the response of system (1) for the initial condition $x(0)$, input u , and attack signal $a \neq 0$. The attack signal a is perfectly undetectable if $y(x(0), u, a) \equiv y(x(0), u, 0)$. ■

Consistent with Milošević et al. (2020), we set $x(0) = 0$ and $u \equiv 0$, in which case $y(0, 0, a) \equiv 0$ for perfect undetectability. This assumption is without loss of generality, since system (1) is linear (Milošević et al., 2020, Section II).

2.2. Security index δ

The security index δ , based on the notion of undetectability as in Definition 1, has been introduced in Milošević et al. (2020). Particularly, the security index $\delta(i)$ of a component i , is defined as follows:

$$\begin{aligned} \delta(i) &:= \underset{a}{\text{minimize}} & \|a\|_0 \\ &\text{subject to} & x(k+1) = Wx(k) + B_a a(k), \quad (C1) \\ & & 0 = Cx(k) + D_a a(k), \quad (C2) \\ & & x(0) = 0, \quad (C3) \\ & & a^{(i)} \neq 0. \quad (C4) \end{aligned}$$

In words, the optimal solution of the problem equals the minimum number of sensors and actuators that ought to be compromised so as to conduct a perfectly undetectable attack. Note that the constraints (C1) and (C2) together ensure that the physical dynamics is in consonance with the model we introduced, while (C2) and (C3) together ensure that the attack signal a is perfectly undetectable. Lastly, constraint (C4) ensures that component i , for which we are computing the index, is actively used in the attack. It can so happen that the aforementioned optimization problem does not have a solution (Milošević et al., 2020). If, for a given component $i \in \mathcal{I}$, the problem of computing $\delta(i)$ does not have a solution, then we set $\delta(i) = +\infty$.

Note that $\delta(i)$ is dependent on the numerical entries of system matrices (W, B_a, C, D_a) , which implies that the value of $\delta(i)$ may change if entries of these matrices change. Furthermore, it is clear that *exact* knowledge of the coefficients of the system matrices (W, B_a, C, D_a) is essential for computation of $\delta(i)$. However, in networked systems such exact knowledge is often not available. Thus, we aim (i) to develop a notion of security index that is impervious to numerical changes in system matrices, and (ii) to obtain bounds on the values that $\delta(i)$ takes across all nonzero values of the coefficients of the system matrices, (W, B_a, C, D_a) . As a first step in this direction, we turn our attention to structured systems.

2.3. Structured system and graphical representations

Let $(W_\omega, B_\beta, C_\gamma, D_\sigma)$ be the structured matrices corresponding to the matrices (W, B_a, C, D_a) , respectively. That is, they have positions that are a priori fixed to zero, and the ones that are not fixed to zero are referred to as free parameters (i.e., are allowed to take any value in $\mathbb{R}$). In other words, the positions corresponding to fixed zeros represent those interactions that are strictly prohibited from occurring. The free parameters model interactions that occur without providing any information regarding the strength of the said interactions. Let $s_\omega, s_\beta, s_\gamma$ and s_σ denote the number of free parameters in $W_\omega, B_\beta, C_\gamma$ and D_σ , respectively, and let $\mathbb{R}^\alpha$, where $\alpha = s_\omega + s_\beta + s_\gamma + s_\sigma$, denote the space of free parameters. Note that each choice of free parameters in $\mathbb{R}^\alpha$ yields a realization of matrices (W, B_a, C, D_a) that determines the system (1).

Let the set of state vertices, actuator vertices, and output vertices be denoted by $X = \{x_1, \dots, x_N\}$, $\mathcal{U} = \{u_1, \dots, u_Q\}$, and

$Y = \{y_1, \dots, y_M\}$, respectively. Define $\mathcal{A} = \{a_{y_j} \mid y_j \in Y \setminus S\}$. That is, $\mathcal{A}$ represents the set of dedicated output attack vertices that can attack the unprotected sensors. The set of all the vertices of the system through which an adversary could potentially inject malicious signals into the system, then, is given by $\mathcal{I} = \mathcal{U} \cup \mathcal{A}$.

Let the edge sets be defined as $\mathcal{E}_\omega = \{(x_j, x_i) \mid [W_\omega]_{ij} \neq 0\}$, $\mathcal{E}_\beta = \{(a_j, x_i) \mid [B_\beta]_{ij} \neq 0\}$, $\mathcal{E}_\gamma = \{(x_j, y_i) \mid [C_\gamma]_{ij} \neq 0\}$, and $\mathcal{E}_\sigma = \{(a_j, y_i) \mid [D_\sigma]_{ij} \neq 0\}$. The graph associated with system (1) is $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = X \cup \mathcal{I} \cup Y$ and $\mathcal{E} = \mathcal{E}_\omega \cup \mathcal{E}_\beta \cup \mathcal{E}_\gamma \cup \mathcal{E}_\sigma$. An illustration of this setup is given in Fig. 1(a), which will hereafter be referred to as **Example 1**. Define the bipartite graph $\mathcal{H} = \{X \cup \mathcal{I}, X' \cup Y, \mathcal{E}\}$. Denote by $\mathcal{E}_{\text{loop}} \subseteq \mathcal{E}$ the existing self-loops in $\mathcal{G}$. Note that the edges in $\mathcal{H}$ are in one-to-one correspondence with the edges of the directed graph $\mathcal{G}$. Let $\mathcal{I}_a \subseteq \mathcal{I}$ denote the (sub)set of actuators and unprotected sensors that are used in an attack. Let $\tilde{\mathcal{H}} \subseteq \mathcal{H}$ be the subgraph of $\mathcal{H}$ formed by considering those vertices of $\mathcal{I}$ that are present in $\mathcal{I}_a$, and the corresponding edges. That is, $\tilde{\mathcal{H}} = \{X \cup \mathcal{I}_a, X' \cup Y, \tilde{\mathcal{E}}\}$. We define yet another bipartite graph $\mathcal{H}_\times = \{X \cup \mathcal{I}, X' \cup Y, \mathcal{E}_\times\}$, where $\mathcal{E}_\times = \mathcal{E} \cup \mathcal{E}_{\text{new}}$. The edge set $\mathcal{E}_{\text{new}}$ represents the newly-added self-loops in the directed graph $\mathcal{G}$. In words, $\mathcal{H}_\times$ is obtained by adding to $\mathcal{H}$ the edge set $\mathcal{E}_{\text{new}}$. Let $\tilde{\mathcal{H}}_\times \subseteq \mathcal{H}_\times$ be a subgraph of $\mathcal{H}_\times$ formed by considering those vertices of $\mathcal{I}$ that are present in $\mathcal{I}_a$, and the corresponding edges. That is, $\tilde{\mathcal{H}}_\times = \{X \cup \mathcal{I}_a, X' \cup Y, \tilde{\mathcal{E}}_\times\}$. Observe that the representations in terms of $\mathcal{G}$ and structured matrices $(W_\omega, B_\beta, C_\gamma, D_\sigma)$ are equivalent. Hence in the sequel, depending on the context, we use the terms “choice of edge weights in $\mathcal{G}$ ” and “choice of free parameters in $\mathbb{R}^\alpha$ ” interchangeably.

Example 1. Consider a networked system whose structural matrices are given as

$$W_\omega = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ w_{21} & 0 & w_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & w_{43} & 0 & 0 \\ 0 & 0 & w_{53} & w_{54} & w_{55} \end{bmatrix},$$

$$B_\beta = \begin{bmatrix} b_{11} & 0 & 0 & 0 \\ b_{21} & b_{22} & 0 & 0 \\ 0 & 0 & b_{33} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$C_\gamma = \begin{bmatrix} c_{11} & 0 & 0 & 0 & 0 \\ 0 & c_{22} & 0 & 0 & 0 \\ 0 & 0 & c_{33} & 0 & 0 \\ 0 & 0 & 0 & c_{44} & c_{45} \end{bmatrix},$$

$$D_\sigma = \begin{bmatrix} 0 & 0 & 0 & d_{14} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then the directed graph associated with this system is as depicted in Fig. 1(a), while the bipartite graphs $\mathcal{H}$ and $\mathcal{H}_\times$ are as shown in Figs. 1(b) and 1(c), respectively.

2.4. Problem statement

In this work, for discrete-time structured LTI systems, we will be interested in developing notions of security indices that are reliant purely on the topology of the underlying graph. More specifically, the main objective of the present paper is to answer the following questions:

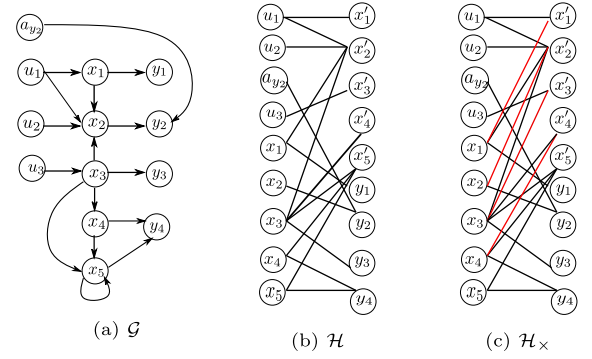


Fig. 1. With respect to **Example 1**, (a): graph $\mathcal{G}$, with $\mathcal{I} = \{u_1, u_2, u_3, a_{y_2}\}$; (b): bipartite graph $\mathcal{H}$; and (c): bipartite graph $\mathcal{H}_\times$, with the edges in red being the newly-added edges. $\mathcal{I} = \{u_1, u_2, u_3, a_{y_2}\}$. $S = \{y_1, y_3, y_4\}$.

- (i) For a given component, $i \in \mathcal{I}$, does $\delta(i)$ have the same value for almost all choices of edge weights in $\mathcal{G}$?
- (ii) Assuming question (i) has a positive answer, how do we compute that value?
- (iii) Third, for a given component, $i \in \mathcal{I}$, what are the graph-theoretic conditions which ensure that for every nonzero choice of edge weights in $\mathcal{G}$, we have $p_1 < \delta(i) \leq p_2$, with $p_1, p_2 \in \mathbb{Z}_+$?

Hereafter, where appropriate, for a given $i \in \mathcal{I}$ we use “ $\delta_s(i)$ ” in place of “ $\delta(i)$ for almost all choices of edge weights in $\mathcal{G}$ ”.

3. Preliminaries

In this section, we acquaint ourselves with notions of ranks of structured matrices, relevant graph-theoretic terminologies, and the interplay between the said notions and graph-theoretic terminologies.

Recall that structured matrices have positions that are a priori fixed to zero and free parameters. Consequently, the rank of such a matrix depends on the choice of free parameters. Given that we are interested in results that are true for almost all (resp. all nonzero) choices of free parameters, we are interested in notions of rank that are independent of the choices of free parameters.

Let $P(z)$ denote the matrix pencil associated with a realization $(W, B_\alpha, C, D_\alpha)$. The normal rank of $P(z)$ (denoted as normal-rank $(P(z))$) is the maximum rank that $P(z)$ attains for some $z_0 \in \mathbb{C}$. Given structured system matrices $(W_\omega, B_\beta, C_\gamma, D_\sigma)$, we define $P_\alpha(z) = \begin{bmatrix} W_\omega - zI_N & B_\beta \\ C_\gamma & D_\sigma \end{bmatrix}$. The generic normal rank of $P_\alpha(z)$ (g-n-rank $P_\alpha(z)$) is the maximum rank that the matrix pencil $P_\alpha(z)$ achieves over all choices of free parameters in $\mathbb{R}^\alpha$ and $z \in \mathbb{C}$ (Sundaram, 2018, Definition C.3).

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of vertices, and $\mathcal{E}$ is the set of edges, we say that two paths in $\mathcal{G}$ are *vertex-disjoint* if the said paths do not share any vertices in common. Let $S_1 \subseteq \mathcal{V}$ and $S_2 \subseteq \mathcal{V}$. A collection of vertex-disjoint paths from S_1 to S_2 is called a *linking* from S_1 to S_2 . Suppose L is a linking in $\mathcal{G}$. If q is one of the vertices along the paths in L , then we say that L *saturates* q . The size of a linking is the number of paths contained in the linking. If a linking L from S_1 to S_2 has maximum size among all the linkings from S_1 to S_2 , then L is a *maximum S_1 - S_2 linking* in $\mathcal{G}$. Let $\mathcal{B}$ be a bipartite graph, with left vertex set V , right vertex set W , and edge set $\mathcal{E}$. A collection of edges $\mathcal{M} \subseteq \mathcal{E}$ form a *matching* if no two edges in $\mathcal{M}$ share a common vertex. If a vertex is present in a matching, we say that the said matching *covers* this vertex. A matching $\mathcal{M}$ of size s is *uniquely restricted* if there exists no other matching $\mathcal{M}' \subseteq \mathcal{E}$ covering exactly the same set of vertices. Given

a bipartite graph $\mathcal{B}$, a uniquely restricted matching, say $\mathcal{M}$, that has the maximum size among all uniquely restricted matchings on the said bipartite graph is referred to as a *maximum uniquely restricted matching*.

It turns out that the generic normal rank of $P_\alpha(z)$ is related to the size of a maximum matching in the bipartite graph $\mathcal{H}_x$, while generic normal rank of the corresponding transfer matrix depends on the maximum size of a linking in $\mathcal{G}$ (van der Woude, 1999; Dion, Commault, & Van Der Woude, 2003). We will be relying on the relevant results in van der Woude (1999), Dion et al. (2003) in the sequel.

4. Main result

In this section, we state and explain the main contributions of the present paper. The detailed analysis of our main results is postponed to the Appendix. The first subsection is concerned with the generic security index of a component and graph-theoretic conditions that aid in its computation. The second subsection provides lower and upper bounds, in graph-theoretic terms, on the value that the security index can take for any nonzero choice of edge weights of $\mathcal{G}$. Hereafter, we use i_1 to indicate the relevant component.

4.1. Generic security index and its computation

In our first main result, we establish that in order to conduct an attack using a particular component, for almost all choices of edge weights in $\mathcal{G}$, the same number of elements is needed.

Theorem 1. For a component $i_1 \in \mathcal{I}$, the security index, $\delta(i_1)$, is generic. ■

Proof. See the Appendix. □

Theorem 1 answers question (i) in Section 2.4, and should be interpreted in the following sense: For almost all choices of free parameters in $\mathbb{R}^\alpha$, the security index of a component i_1 , $\delta(i_1)$, remains the same (i.e., p). For some choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1)$ could be different from p . However, all such choices of free parameters would lie on a set of Lebesgue measure zero in the space of free parameters.

The following theorem establishes graph-theoretic conditions for computing the generic security index of a component i_1 :

Theorem 2. Let $p \in \mathbb{Z}_+$ and $1 \leq p \leq |\mathcal{I}|$. Then

- (i) If for every $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $i_1 \in \mathcal{I}_a$, and b) $|\mathcal{I}_a| = p$, i_1 is saturated by every maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$, then $\delta_s(i_1) \geq p + 1$.
- (ii) If for some $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $i_1 \in \mathcal{I}_a$, and b) $|\mathcal{I}_a| = p$, there exists a maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$ that does not saturate i_1 , then $\delta_s(i_1) \leq p$. ■

Proof. See the Appendix. □

Theorem 2 should be interpreted as follows: if, for some $p \in \mathbb{Z}_+$, condition in item (i) is satisfied, and, for $p + 1$ condition in item (ii) is satisfied (or, equivalently, condition in item (i) is violated) then for almost all choices of edge weights in $\mathcal{G}$, $\delta(i_1) = p$. Theorem 2 thus enables the computation of the generic security index, thereby answering question (ii) in Section 2.4.

Thanks to Commault, Dion, and van der Woude (2002, Lemma 4), we can rewrite the conditions in Theorem 2 in terms of matchings of appropriate size on the bipartite graph $\mathcal{H}_x$. In order to do so, we introduce the relevant subgraph of $\mathcal{H}_x$, namely $\tilde{\mathcal{H}}_{x-i_1}$ obtained by removing vertex i_1 and the corresponding edges from $\mathcal{H}_x$. We have the following theorem.

Theorem 3. Let $p \in \mathbb{Z}_+$ such that $1 \leq p \leq |\mathcal{I}|$. Let ℓ be the number of sets $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$. For $k = 1, 2, \dots, \ell$, let $\tilde{\mathcal{H}}_x^k$ denote the bipartite graph corresponding to set $\mathcal{I}_a^k$.

- (i) If for $k = 1, 2, \dots, \ell$, the size of a maximum matching $\tilde{\mathcal{M}}_x^k$ in $\tilde{\mathcal{H}}_x^k$ is strictly less than the size of a maximum matching $\tilde{\mathcal{M}}_x^k$ in $\tilde{\mathcal{H}}_{x-i_1}^k$, then $\delta_s(i_1) \geq p + 1$.
- (ii) If for some $k \in \{1, 2, \dots, \ell\}$, the size of a maximum matching $\tilde{\mathcal{M}}_x^k$ in $\tilde{\mathcal{H}}_x^k$ is equal to the size of a maximum matching $\tilde{\mathcal{M}}_x^k$ in $\tilde{\mathcal{H}}_{x-i_1}^k$, then $\delta_s(i_1) \leq p$. ■

Proof. See the Appendix. □

Remark 1. If condition in item (i) of Theorem 3 is satisfied for $p = |\mathcal{I}|$, then for almost all choices of edge weights in $\mathcal{G}$ the attacker cannot attack i_1 while staying perfectly undetectable. Hence, $\delta_s(i_1) = +\infty$. ■

Using Commault et al. (2002, Lemma 4) and arguments similar to that in Lemma 1 (see Appendix), the conditions in Theorem 3, for a given set $\mathcal{I}_a \subseteq \mathcal{I}$, could be rephrased in terms of whether (or not) vertex i_1 is covered by all maximum matchings in $\tilde{\mathcal{H}}_x$. We now introduce a brute force search method for calculating the generic security index $\delta_s(i_1)$ of a component i_1 ; see Algorithm 1. The idea is to iterate through all subsets $\mathcal{I}_a$ that contain i_1 and have cardinality p , $1 \leq p \leq |\mathcal{I}|$, starting from $p = 1$. If we find a subset for which condition (ii) in Theorem 3 is satisfied, then $\delta_s(i_1) = p$, and we stop the search. Otherwise, we increment p , and repeat the process. It turns out that by following the aforementioned steps, we are guaranteed to compute the exact value of $\delta_s(i_1)$ for every component $i_1 \in \mathcal{I}$, as established in the following proposition:

Proposition 1 (Correctness of Algorithm 1). Let $i_1 \in \mathcal{I}$ be an arbitrary component. The value returned by Algorithm 1 is equal to $\delta_s(i_1)$. ■

Proof. See the Appendix. □

Note that by leveraging the conditions in Theorem 2 Algorithm 1 could also be rewritten in terms of linkings on the directed graph $\mathcal{G}$. The details are omitted in the interest of space.

Algorithm 1 Computing $\delta_s(i_1)$.

Input: $W_\omega, B_\beta, C_\gamma, D_\sigma, i_1 \in \mathcal{I}$

Output: $\delta_s(i_1)$

- 1: Set $p = 1$
- 2: Generate all subsets $\mathcal{I}_a$ of $\mathcal{I}$ such that: (i) $|\mathcal{I}_a| = p$; (ii) $i_1 \in \mathcal{I}_a$; and calculate ℓ , the number of such subsets.
- 3: for $j = 1 : \ell$
- 4: compute all the maximum matchings in the bipartite graph $\tilde{\mathcal{H}}_x^j$
- 5: If there exists a maximum matching which does not cover i_1 , then return $\delta_s(i_1) = p$. and terminate the algorithm.
- 6: Otherwise, if $p < |\mathcal{I}|$, then set $p = p + 1$, and return to Line 2. Else return $\delta_s(i_1) = +\infty$, and terminate the algorithm.

Example 1 (contd.). It turns out that for the example in Fig. 1(a), $\delta_s(u_2) = 2$, $\delta_s(a_{y_1}) = 2$, $\delta_s(u_1) = +\infty$, and $\delta_s(u_3) = +\infty$; please see Gracy et al. (2021, Section 1.1) for details.

Please note that we provide another example to illustrate the effectiveness of Algorithm 1 for a non-trivial system; please see Gracy et al. (2021, Section 2.2) for details.

Note that if $\delta_s(i) = p$, then Algorithm 1 generates all the subsets of $\mathcal{I}$ whose cardinality is less than or equal to p , and

checks condition (ii) in Theorem 2 for each of these subsets. Thus, if $|\mathcal{I}|$ and p are large, then the algorithm can become prohibitively time consuming for systems containing hundreds of components. Yet, two properties of this algorithm can allow us to compute $\delta_s(i_1)$ for a large class of systems. First, the maximum matching on a bipartite graph can be computed using the Hopcroft–Karp maximum matching algorithm, which has polynomial-time complexity (Hopcroft & Karp, 1973). Second, checking condition (ii) can be done independently for every subset we generate. Hence, the execution time of Algorithm 1 can significantly be improved by using parallel computing. These two properties allow us to efficiently compute the generic security index for realistic systems. We highlight the same using an example in Gracy et al. (2021, Section 2.2).

Remark 2 (Generic Security Index and Attacks). The following example shows how it is possible to design a perfectly undetectable attack on a component compromising a set of other components of the system. Consider the system given by

$$\begin{aligned} x(k+1) &= \begin{bmatrix} 0 & w_{12} \\ 0 & 0 \end{bmatrix} x(k) + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} a(k), \\ y(k) &= \begin{bmatrix} c_{11} & 0 \\ c_{21} & 0 \end{bmatrix} x(k) + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} a(k). \end{aligned}$$

This system has two states, x_1 and x_2 , two actuators, u_1 and u_2 , and two outputs y_1 and y_2 . Both the actuators can be corrupted by an adversary. Further, we assume that both the outputs can be attacked, and hence we have $a(k) = [u_1(k) \ u_2(k) \ a_{y_1}(k) \ a_{y_2}(k)]^T$. Using Algorithm 1, we see that $\delta_s(u_1) = 2$. Namely, say the attacker launches an arbitrary attack $a_1(k)$ against u_1 . She can then make this attack perfectly undetectable in almost all realization by attacking actuator u_2 with the signal $a_2(k) = -a_1(k+1)/w_{12}$. ■

Given a graph $\mathcal{G}$, both Theorems 2 and 3 give conditions to compute the generic security index for every vulnerable component, i.e., for every element in $\mathcal{I}$. However, there might be scenarios where a particularly crucial component, say $i_1 \in \mathcal{I}$, has a low generic security index. In such a context, the system operator would like to increase the generic security index of i_1 . Towards this end, the graph $\mathcal{G}$ has to be modified; in the sense that more edges have to be added, although the number of nodes remain unchanged. We refer to this newly-obtained graph as $\hat{\mathcal{G}}$. The following corollary shows how such a modification indeed achieves the desired objective.

Corollary 1. Given a graph $\mathcal{G}$, and a component $i_1 \in \mathcal{I}$ such that $\delta_s(i_1) = p$. Then if edges are added to $\mathcal{G}$ such that for every $\mathcal{I}_a \in \mathcal{I}$ that satisfies $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$, there exists a linking of size $|\mathcal{I}_a|$ from $\mathcal{I}_a$ to $\mathcal{Y}$ in $\hat{\mathcal{G}}$, then $\delta_s(i_1) \geq p + 1$. ■

Proof. See the Appendix. □

With respect to Example 1 (see Fig. 1(a)), we aim to boost the generic security index of u_2 such that $\delta_s(u_2) \geq 3$. To this end, we suitably modify graph $\mathcal{G}$ by adding extra edges, i.e., $\hat{\mathcal{G}}$; see Fig. 2(c). Using Algorithm 1, but taking the input as $\hat{\mathcal{G}}$, instead of $\mathcal{G}$, we see that $\delta_s(u_2) \geq 3$.

The notion of generic security index gives a measure of the vulnerability of a component, and this measure is robust to variations in system parameters. However, it suffers from the following limitation: With respect to Example 1, consider a choice of free parameters that yields the following realization:

$$W = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 4 & 3 \end{bmatrix}, \quad B_a = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 5 & 3 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

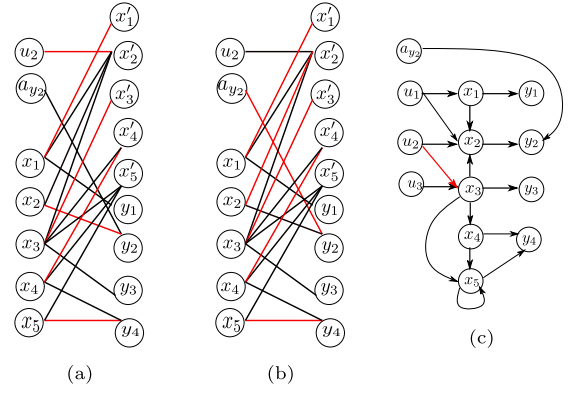


Fig. 2. (a): bipartite graph $\tilde{\mathcal{H}}_x$ corresponding to set $\mathcal{I}_a = \{a_{y_2}, u_2\}$, with edges in red forming a matching of size 6, but not covering a_{y_2} ; (b): bipartite graph $\tilde{\mathcal{H}}_x$ corresponding to set $\mathcal{I}_a = \{a_{y_2}, u_2\}$, with edges in red forming a matching of size 5, but not covering u_2 ; and (c): With respect to Example 1, graph $\hat{\mathcal{G}}$. The edges in red denote the extra edges added to $\mathcal{G}$, while the edges in black are the same as those in $\mathcal{G}$.

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 3 & 6 \end{bmatrix}, \quad D_a = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (2)$$

We ask the following: “Given that $\delta_s(a_{y_2}) = 2$, with respect to the aforementioned realization, what is $\delta(a_{y_2})$?” The answer is $\delta(a_{y_2})$ might (or might not) be equal to two, since the said realization could well lie on the Lebesgue set of measure zero where $\delta(a_{y_2})$ is not necessarily 2. Hence, assuming certain realizations occur frequently enough, and these realizations lie on the Lebesgue set of measure zero where $\delta(i_1)$ is not necessarily p , $\delta_s(i_1)$ may result in an inaccurate measure of security of component i_1 . Thus, it is clear that for applications/settings where a perfectly undetectable attack can result in significant impacts, the guarantees provided by $\delta_s(i_1)$ may not suffice. Thus, in order to obtain stronger (i.e., deterministic) guarantees of security of a component, it would be prudent to find upper (resp. lower) bound on $\delta(i_1)$ that accounts for all nonzero choices of edge weights in $\mathcal{G}$. We will address this problem next.

4.2. Bounds on $\delta(i_1)$ for all nonzero choices of edge weights of $\mathcal{G}$

We are interested in finding graph-theoretic conditions for computing the upper (resp. lower) bounds on the security index of component i_1 , i.e., $\delta(i_1)$, for all nonzero choices of edge weights in $\mathcal{G}$ or, equivalently, for all nonzero choices of free parameters in $\mathbb{R}^a$. In order to obtain these graph-theoretic conditions, we introduce the relevant subgraph of $\tilde{\mathcal{H}}$, namely $\tilde{\mathcal{H}}_{-i_1}$ obtained by removing vertex i_1 and the corresponding edges from $\tilde{\mathcal{H}}$. We have the following results:

Theorem 4. Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$. If

- (i) the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}$ in $\tilde{\mathcal{H}}$ equals the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}$ in $\tilde{\mathcal{H}}_{-i_1}$, and
- (ii) the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}_x$ in $\tilde{\mathcal{H}}_x$ such that $\tilde{\mathcal{M}}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$ equals the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}_x$ in $\tilde{\mathcal{H}}_{x-i_1}$ such that $\tilde{\mathcal{M}}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$

then, for all nonzero choices of edge weights of $\mathcal{G}$, $\delta(i_1) \leq p$. ■

Proof. See the Appendix. □

In words, if the conditions in [Theorem 4](#) are satisfied, then a perfectly undetectable attack involving i_1 , can be conducted for every nonzero choice of edge weights in $\mathcal{G}$, using p elements.

Theorem 5. Let $p \in \mathbb{Z}_+$ such that $1 \leq p \leq |\mathcal{I}|$. Let ℓ be the number of sets $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$. For $k = 1, 2, \dots, \ell$, let $\tilde{\mathcal{H}}^k$ (resp. $\tilde{\mathcal{H}}_{\times}^k$) denote the bipartite graph corresponding to set $\mathcal{I}_a^k$. If for $k = 1, 2, \dots, \ell$

- (i) the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}^k$ in $\tilde{\mathcal{H}}^k$ is strictly less than the maximum size of a uniquely restricted matching $\hat{\mathcal{M}}^k$ in $\tilde{\mathcal{H}}_{-i_1}^k$, and
- (ii) the maximum size of a uniquely restricted matching $\tilde{\mathcal{M}}_{\times}^k$ in $\tilde{\mathcal{H}}_{\times}^k$ such that $\tilde{\mathcal{M}}_{\times}^k \cap \mathcal{E}_{\text{loop}} = \emptyset$ is strictly less than the maximum size of a uniquely restricted matching $\hat{\mathcal{M}}_{\times}^k$ in $\tilde{\mathcal{H}}_{\times-i_1}^k$ such that $\hat{\mathcal{M}}_{\times}^k \cap \mathcal{E}_{\text{loop}} = \emptyset$

then, for all nonzero choices of edge weights of $\mathcal{G}$, $\delta(i_1) \geq p + 1$. ■

Proof. See the [Appendix](#). □

In words, if the conditions in [Theorem 5](#) are satisfied, then, for every nonzero choice of edge weights, in order to conduct a perfectly undetectable attack involving i_1 , at least $p + 1$ elements are needed. Note that [Theorems 4](#) and [5](#) together aid in addressing question (iii) in [Section 2.4](#).

The understanding behind the bounds introduced in the previous theorems is along the following lines. First, using the conditions in [Theorem 4](#), if the upper bound computed is too low, then component i_1 can be attacked with a small number of resources in any nonzero realization. This further implies that component i_1 is extremely vulnerable. However, a high upper bound does not necessarily mean that component i_1 is secure. Second, using the conditions in [Theorem 5](#), if the lower bound computed is too high, then it is clear that component i_1 is extremely secure. However, a low lower bound does not necessarily mean that component i_1 is quite vulnerable.

Next, we discuss the problem of computing these bounds. While, with respect to a given set, the graph theoretic condition for computing $\delta_s(i_1)$ can be checked efficiently, this is not the case with the conditions that characterize the bounds for obtaining deterministic guarantees. The following proposition establishes the same.

Proposition 2. Checking the conditions in [Theorems 4](#), or the conditions in [Theorem 5](#) is NP-hard. ■

Proof. See the [Appendix](#). □

Nevertheless, these conditions can still be verified in small scale systems by generating all the edge combinations and checking if the conditions from [Theorems 4](#) and [5](#) are satisfied, as we will illustrate in [Example 1](#). As with computing $\delta_s(i)$, we can use parallelization to improve the running time of such a search. Additionally, it turns out that one can find conditions, less restrictive than those in [Theorem 5](#).

Corollary 2. Let $p \in \mathbb{Z}_+$ such that $1 \leq p \leq |\mathcal{I}|$. Let ℓ be the number of sets $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$. For $k = 1, 2, \dots, \ell$, let $\tilde{\mathcal{H}}^k$ (resp. $\tilde{\mathcal{H}}_{\times}^k$) denote the bipartite graph corresponding to set $\mathcal{I}_a^k$. If

- (i) either for every bipartite graph $\tilde{\mathcal{H}}^k$, $k = 1, 2, \dots, \ell$, there exists a uniquely restricted matching $\tilde{\mathcal{M}}^k$ of size $N + p$ in $\tilde{\mathcal{H}}^k$, or
- (ii) for every bipartite graph $\tilde{\mathcal{H}}_{\times}^k$, $k = 1, 2, \dots, \ell$, there exists a uniquely restricted matching $\tilde{\mathcal{M}}_{\times}^k$ of size $N + p$ in $\tilde{\mathcal{H}}_{\times}^k$,

then, for all nonzero choices of edge weights of $\mathcal{G}$, $\delta(i_1) \geq p + 1$. ■

Proof. See the [Appendix](#). □

The conditions in [Corollary 2](#), with respect to a given candidate set $\mathcal{I}_a$, essentially ask the following: In a given bipartite graph, is there a maximum matching of a given size which is simultaneously uniquely restricted? An algorithm to answer this question in polynomial time is known ([Penso, Rautenbach, & dos Santos Souza, 2015](#), Corollary 4). Moreover, since the size we look for equals the size of the left vertex set, a simpler algorithm can also be used ([Chapman & Mesbahi, 2013](#); [Hartung, Reißig, & Svaricek, 2012](#)). However, if the conditions from [Corollary 2](#) do not hold, then one needs to check the conditions from [Theorem 5](#).

Next, we explain how to calculate the bounds on the values that $\delta(i_1)$ achieves for all nonzero choices of free parameters in $\mathbb{R}^{\alpha}$. Set $p = 1$. Generate all subsets $\mathcal{I}_a \subseteq \mathcal{I}$ of cardinality p that contain i_1 . For each of such subsets, check if condition (i) or condition (ii) in [Corollary 2](#) is satisfied. If yes, then increment p . If, for some subset, neither of the conditions in [Corollary 2](#) is satisfied, then check whether both the conditions in [Theorem 5](#) are satisfied. If yes, then increment p . If no, then, for the said subset, check whether or not both the conditions in [Theorem 4](#) is satisfied. If yes, then, for all nonzero choices of free parameters in $\mathbb{R}^{\alpha}$, $\delta(i_1) \leq p$. If not, then, for all nonzero choices of free parameters in $\mathbb{R}^{\alpha}$, $\delta(i_1) \geq \mu$, where μ is the last known value of p for which at least one of the following happens: a) either both the conditions in [Corollary 2](#) is satisfied, or b) both the conditions in [Theorem 5](#) is satisfied. If, for $p = |\mathcal{I}|$, the conditions in [Corollary 2](#) (resp. [Theorem 5](#)) are satisfied, then, consistent with [Remark 1](#), we conclude that, for all nonzero choices of free parameters in $\mathbb{R}^{\alpha}$, $\delta(i_1) = +\infty$.

Example 1 (contd.): It turns out that for all nonzero choices of edge weights of the graph in [Fig. 1\(a\)](#), (i) $\delta(a_{y_2}) \leq 2$, and $\delta(a_{y_2}) \geq 2$, thus implying $\delta(a_{y_2}) = 2$, (ii) $\delta(u_2) \geq 2$, and (iii) $\delta(u_3) = +\infty$. Please see [Gracy et al. \(2021, Section 1.2\)](#) for details.

Note that since, for all nonzero choices of edge weights, $\delta(a_{y_2}) = 2$, it follows that, with respect to the realization given in [\(2\)](#), $\delta(a_{y_2}) = 2$.

For an illustration of our theoretical findings in a more realistic setting, please see [Gracy et al. \(2021, Section 2.1\)](#)

5. Conclusion

In this paper, for discrete-time LTI network systems represented by a directed graph, we showed that the security index remains the same for almost all choices of edge weights of the said graph. Thereafter, we provided graph-theoretic conditions in terms of both linkings (on directed graphs) and matchings (on appropriately defined bipartite graphs) that aid in computing the generic security index of a component. Next, we established graph-theoretic conditions, in terms of uniquely restricted matchings on suitably-defined bipartite (sub)graphs, that yield lower (resp. upper) bounds on the values that the security index can take for all nonzero choices of edge weights of the said graph. We then showed that computing the said bounds is NP-hard. Finally, we illustrated our theoretical findings through two practical examples, the details of which are available in [Gracy et al. \(2021\)](#).

Appendix

Let us denote by $(W, \tilde{B}_a, C, \tilde{D}_a)$ the corresponding system, where $\begin{bmatrix} \tilde{B}_a \\ \tilde{D}_a \end{bmatrix}$ the submatrix of $\begin{bmatrix} B_a \\ D_a \end{bmatrix}$ formed by looking at columns of $\begin{bmatrix} B_a \\ D_a \end{bmatrix}$ that correspond to elements in $\mathcal{I}_a$. Denote by $G_{\alpha}^{\mathcal{I}_a}(z)$ the transfer matrix of the corresponding structured system.

We will need the following lemmas for proving [Theorem 1](#).

Lemma 1. Let $\mathcal{I}_a \subseteq \mathcal{I}$. If for some $i \in \mathcal{I}_a$, $\text{g-n-rank } G^{\mathcal{I}_a} = \text{g-n-rank } G^{\mathcal{I}_a \setminus i} + 1$, then vertex i is saturated by every maximum linking from $\mathcal{I}_a$ to Y . ■

Proof. Let $\text{g-n-rank } G^{\mathcal{I}_a} = r$, where $1 \leq r \leq |\mathcal{I}_a|$. Then, from [Dion et al. \(2003, Theorem 2\)](#), the size of a maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$ equals r . Suppose that vertex i is not saturated by some maximum linking, say L , from $\mathcal{I}_a$ to Y in $\mathcal{G}$. This implies that the size of maximum linking, namely size of linking L , from $\mathcal{I}_a \setminus i$ to Y equals r , since if it were $r - 1$ then L should have included vertex i . Consequently, $\text{g-n-rank } G^{\mathcal{I}_a \setminus i} = r$, and hence, $\text{g-n-rank } G^{\mathcal{I}_a} \neq \text{g-n-rank } G^{\mathcal{I}_a \setminus i} + 1$. □

Lemma 2. Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$. If every maximum linking from $\mathcal{I}_a$ to Y saturates i_1 , then using $\mathcal{I}_a$, for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, an attacker cannot attack i_1 and remain perfectly undetectable. ■

Proof. Let $r \in \mathbb{Z}_+$ be the size of a maximum linking from $\mathcal{I}_a$ to Y in the directed graph $\mathcal{G}$. Denote by $G^{\mathcal{I}_a}(z)$ the transfer matrix of the corresponding structured system.

Pick a maximum linking, say L_1 , from $\mathcal{I}_a$ to Y in the directed graph $\mathcal{G}$, and do the following: for all edges of $\mathcal{G}$ that are contained in L_1 , set the weights to 1; for the rest, set the weights to 0. As a consequence of this choice of edge weights, and since size of a maximum linking from $\mathcal{I}_a$ to Y in the directed graph $\mathcal{G}$ equals r , from the definition of linking it follows that for the aforementioned choice of free parameters, $\text{rank } G^{\mathcal{I}_a}(z) = r$. Moreover, column i_1 is linearly independent of the remaining columns in $G^{\mathcal{I}_a}(z)$. Since there exists one realization for which $\text{rank } G^{\mathcal{I}_a}(z) = r$, then for almost all choices of edge weights in $\mathcal{G}$ $\text{rank } G^{\mathcal{I}_a}(z) = r$, and all such realizations would have column i_1 to be linearly independent of the remaining columns in $G^{\mathcal{I}_a}(z)$.

Given that every maximum linking from $\mathcal{I}_a$ to Y saturates i_1 , it follows that, even though repeating the procedure above for different maximum linkings in $\mathcal{G}$ would yield different $G^{\mathcal{I}_a}(z)$, for all such transfer matrices of the corresponding structured systems, $G^{\mathcal{I}_a}(z)$, the following are satisfied: a) $\text{g-n-rank } G^{\mathcal{I}_a}(z) = r$, and b) for almost all realizations, column i_1 will be linearly independent of the remaining columns in $G^{\mathcal{I}_a}(z)$, $i \neq 1$.

Therefore, for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the following is satisfied: let $x(0) = 0$, $u = 0$, then $y = 0 \implies a^{(i_1)} = 0$, which, with respect to set $\mathcal{I}_a$, is in violation of constraint (C4). Hence, using $\mathcal{I}_a$, for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, an attacker cannot attack i_1 and remain perfectly undetectable. □

We need the following lemma to establish the result in [Theorem 2](#).

Lemma 3. Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$, and let $\mathcal{G}$ be the associated directed (sub)graph. Vertex i_1 is saturated by every maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$ if and only if $\text{g-n-rank } G^{\mathcal{I}_a} = \text{g-n-rank } G^{\mathcal{I}_a \setminus i_1} + 1$. ■

Proof. Let r , where $1 \leq r \leq |\mathcal{I}_a|$ be the size of a maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$. Then, from [Dion et al. \(2003, Theorem 2\)](#), $\text{g-n-rank } G^{\mathcal{I}_a} = r$. Suppose i_1 is saturated by every maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}$, then removing vertex i_1 from $\mathcal{I}_a$ should reduce the size of the maximum linking from $\mathcal{I}_a \setminus i_1$ to Y in $\mathcal{G}$ by one, because otherwise there would be a maximum linking from $\mathcal{I}_a$ to Y in $\mathcal{G}_F$ that does not saturate i_1 . Hence, the maximum size of a linking from $\mathcal{I}_a \setminus i_1$ to Y equals $r - 1$. This further implies that $\text{g-n-rank } G^{\mathcal{I}_a \setminus i_1} = r - 1$.

The other direction is handled in [Lemma 1](#). □

We will need the following propositions for proving [Theorem 1](#).

Proposition 3. If for every $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$, $\text{g-n-rank } G^{\mathcal{I}_a} \neq \text{g-n-rank } G^{\mathcal{I}_a \setminus i_1}$, then $\delta_s(i_1) \geq p + 1$. ■

Proof. Suppose that the conditions in [Proposition 3](#) are satisfied, yet $\delta_s(i_1) \leq p$. That is, there exists some $\mathcal{I}_a^1 \subseteq \mathcal{I}$, where a) $|\mathcal{I}_a^1| = p$, and b) $i_1 \in \mathcal{I}_a^1$, such that for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, where $[\tilde{B}_\beta]_{\tilde{D}_\sigma}$ is the structured submatrix of $[\tilde{B}_\beta]$ with column indices corresponding to elements in $\mathcal{I}_a^1$, the following is satisfied: let $x(0) = 0$, $u = 0$, then $y = 0 \implies a^{(i_1)} \neq 0$.

Let $r \in \mathbb{Z}_+$, where $1 \leq r \leq |\mathcal{I}_a^1| - 1$, denote the size of a maximum linking from $\mathcal{I}_a^1$ to Y in $\mathcal{G}$. Then, from [Dion et al. \(2003, Theorem 2\)](#), $\text{g-n-rank } G^{\mathcal{I}_a^1} = r$. Hence, $\text{g-n-rank } G^{\mathcal{I}_a^1 \setminus i_1} \geq r - 1$. Now there are two cases to consider here.

Case a: If $\text{g-n-rank } G^{\mathcal{I}_a^1 \setminus i_1} = r$, then, by contraposition, the proof is complete.

Case b: If $\text{g-n-rank } G^{\mathcal{I}_a^1 \setminus i_1} = r - 1$, then $\text{g-n-rank } G^{\mathcal{I}_a^1} = \text{g-n-rank } G^{\mathcal{I}_a^1 \setminus i_1} + 1$. Hence, from [Lemma 1](#), vertex i_1 is saturated by every maximum linking from $\mathcal{I}_a^1$ to Y . This implies, from [Lemma 2](#), that for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, an attacker cannot attack i_1 and remain perfectly undetectable. This contradicts our initial assumption. Thus, the proof is complete. □

Proposition 4. Let $\mathcal{I}_a \subseteq \mathcal{I}$, and $i_1 \in \mathcal{I}_a$. If $\text{g-n-rank } G^{\mathcal{I}_a} = \text{g-n-rank } G^{\mathcal{I}_a \setminus i_1}$, then $\delta_s(i_1) \leq |\mathcal{I}_a|$. ■

Proof. Let us denote by $[\tilde{B}_\beta]_{\tilde{D}_\sigma}$ the columns of $[\tilde{B}_\beta]$ that correspond to elements in $\mathcal{I}_a$. By assumption, all such structured submatrices $[\tilde{B}_\beta]_{\tilde{D}_\sigma}$ contain column i_1 . Let us denote by $(W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$ the corresponding structured subsystem of $(W_\omega, B_\beta, C_\gamma, D_\sigma)$. Let $G^{\mathcal{I}_a}$ denote the transfer matrix of the corresponding structured system, and let $\text{g-n-rank } G^{\mathcal{I}_a} = r$, where $1 \leq r \leq |\mathcal{I}_a|$.

Suppose that $\delta_s(i_1) > |\mathcal{I}_a|$. Then, together with the definition of security indices of a component, it follows that for every set of size $|\mathcal{I}_a|$ containing i_1 , for almost all realizations $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the following is satisfied: let $x(0) = 0$, $u = 0$, then $y = 0 \implies a^{(i_1)} = 0$. This, since $G^{\mathcal{I}_a} = [G^{\mathcal{I}_a \setminus \{i_1\}} \quad G^{(i_1)}]$, implies that $\text{g-n-rank } G^{(i_1)} = 1$, and moreover, column i_1 is linearly independent of the columns of $G^{\mathcal{I}_a \setminus \{i_1\}}$. Therefore, $\text{g-n-rank } G^{\mathcal{I}_a \setminus \{i_1\}} = r - 1$, which further implies that $\text{g-n-rank } G^{\mathcal{I}_a} \neq \text{g-n-rank } G^{\mathcal{I}_a \setminus \{i_1\}}$. This completes the proof. □

Proposition 5. Let $p \in \mathbb{Z}_+$ such that $1 \leq p \leq |\mathcal{I}|$.

- (i) If for every $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$, $\text{g-n-rank } G^{\mathcal{I}_a} \neq \text{g-n-rank } G^{\mathcal{I}_a \setminus i_1}$, then $\delta_s(i_1) \geq p + 1$.
- (ii) If $\text{g-n-rank } G^{\mathcal{I}_a} = \text{g-n-rank } G^{\mathcal{I}_a \setminus i_1}$, then $\delta_s(i_1) \leq p$. ■

Proof. The proof follows by combining the results in [Proposition 3](#) and [4](#). □

Proof of Theorem 1. Indeed, it turns out that the security index of a component is generic. To see this, consider the following argument: First note that the conditions in [Proposition 5](#) are mutually exclusive. For $p = 1$, it follows from the definition of security index that the condition in item (i) is trivially satisfied for every $i \in \mathcal{I}$. Therefore, one needs to check for $p \neq 1$, starting from $p = 2$. This is done as follows: generate all sets $\mathcal{I}_a$ such that $i \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$. If the condition in item (i) is satisfied for

every such set $\mathcal{I}_a$, then at least $p + 1$ elements are needed. Let $\mathcal{I}_a$ be the first set for which the condition in item (ii) (resp. item (i)) is satisfied (resp. violated). Then a perfectly undetectable attack using component i can be conducted with at most p elements for almost all choices of edge weights in $\mathcal{G}$. However, since the condition was satisfied for $p = p - 1$, it follows that in order to conduct an attack using component i exactly p elements are needed for almost all choices of edge weights in $\mathcal{G}$. This implies that $\delta_s(i) = p$, thus ending the proof. $\square$

Proof of Theorem 2. From Proposition 5 and Lemma 3, the result in Theorem 2 is readily obtained, thereby completing the proof. $\square$

Proof of Theorem 3. The proof follows immediately from Proposition 5, and by exploiting the connection between generic normal rank of a transfer function matrix and the size of a maximum matching on a suitably-defined bipartite graph, as given in Commault et al. (2002, Lemma 4). $\square$

Proof of Proposition 1. First notice that, starting from $p = 1$, step 1 ensures that, for a given value of p , every (sub)set of p components, $\mathcal{I}_a$, necessarily includes component i_1 . For a given (sub)set $\mathcal{I}_a$, steps 2 and 3 together check whether (or not) component i_1 is covered by every maximum matching on the corresponding bipartite graph $\tilde{\mathcal{H}}_\times$. Notice that the conditions in Theorem 3 are mutually exclusive, and therefore every (sub)set $\mathcal{I}_a$ must satisfy one, and exactly one, of the items in step 3. If the answer for the aforementioned checking is negative, then, from condition (ii) in Theorem 3, $\delta_s(i_1) \leq p$. Since condition (ii) in Theorem 3 was not satisfied for $p - 1$ (because otherwise $\delta_s(i_1) \leq p - 1$, and, therefore, there is no need to check for p), it follows that $\delta_s(i_1) = p$. If, for a given value of p , all the subsets $\mathcal{I}_a$ (as described in i) and ii) of Step 1) yield positive answer in step 3, then, it follows from condition (i) in Theorem 3 that $\delta_s(i_1) \geq p$. Therefore, in step 4 we increment p , provided $p \leq |\mathcal{I}|$, and repeat steps 1 – 3. If, on the contrary, $p > |\mathcal{I}|$, then it is clear that using the elements in set $\mathcal{I}$, for almost all choices of free parameters in $\mathbb{R}^\alpha$, it is not possible to conduct a perfectly undetectable attack using component i , and therefore, consistent with Remark 1, we have $\delta_s(i_1) = +\infty$. $\square$

Proof of Corollary 1. Suppose that the condition is satisfied. Then for every set $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$, from Dion et al. (2003, Theorem 2) it follows that $\text{g-n-rank } G^{\mathcal{I}_a} = p$. Since $|\mathcal{I}_a| = p$, it is immediate that $\text{g-n-rank } G^{\mathcal{I}_a \setminus \{i_1\}} = p - 1$, and, therefore, $\text{g-n-rank } G^{\mathcal{I}_a} = \text{g-n-rank } G^{\mathcal{I}_a \setminus \{i_1\}}$. This, from Proposition 3, further implies that $\delta_s(i_1) \geq p$. $\square$

We need the following lemmas for proving Theorems 4 and 5

Lemma 4. Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}_a$ and $|\mathcal{I}_a| = p$, and let $(W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$ be the corresponding structured subsystem. If

- (i) there exists a uniquely restricted matching $\tilde{\mathcal{M}}$ of size r on the bipartite graph $\tilde{\mathcal{H}}$, and
- (ii) there exists a uniquely restricted matching $\mathcal{M}_\times$ of size r on the bipartite graph $\tilde{\mathcal{H}}_\times$ such that $\tilde{\mathcal{M}}_\times \cap \mathcal{E}_{\text{loop}} = \emptyset$

then for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the corresponding matrix pencil, $P^{\mathcal{I}_a}(z)$, satisfies $\text{rank } P^{\mathcal{I}_a}(z) = r, \forall z \in \mathbb{C}$. $\blacksquare$

Proof. The proof is quite similar to that of Gracy, Garin, and Kibangou (2019, Theorem 1), and is skipped in the interest of space. $\square$

Lemma 5. If $\text{rank } P^{\mathcal{I}_a}(z) = \text{rank } P^{\mathcal{I}_a \setminus i_1}(z)$ for all $z \in \mathbb{C}$, then $\text{normal-rank } P^{\mathcal{I}_a}(z) = \text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z)$. $\blacksquare$

Proof. Note that for a given matrix pencil $P^{\mathcal{I}_a}(z)$, for all $z \in \mathbb{C}$, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) \leq \text{normal-rank } P^{\mathcal{I}_a}(z)$. To see this, consider the following (contradiction) argument: Suppose that there exists some $z_0 \in \mathbb{C}$ such that, for all $z \in \mathbb{C}$, $\text{normal-rank } P^{\mathcal{I}_a}(z) < \text{rank } P^{\mathcal{I}_a \setminus i_1}(z_0)$. This, from the definition of normal rank, implies $\text{rank } P^{\mathcal{I}_a}(z_0) < \text{rank } P^{\mathcal{I}_a \setminus i_1}(z_0)$. However, for any matrix pencil (in particular, $P^{\mathcal{I}_a \setminus i_1}(z)$), addition of a column (namely, i_1) does not decrease the rank of $P^{\mathcal{I}_a}(z)$, which contradicts our initial assumption, thus establishing the claim.

Since for all $z \in \mathbb{C}$, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) \leq \text{normal-rank } P^{\mathcal{I}_a}(z)$, it follows that $\text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z) \leq \text{normal-rank } P^{\mathcal{I}_a}(z)$. By way of contraposition, assume that $\text{normal-rank } P^{\mathcal{I}_a}(z) \neq \text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z)$. This implies that $\text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{normal-rank } P^{\mathcal{I}_a}(z)$. Suppose that $P^{\mathcal{I}_a}(z)$ achieves maximum rank at some $z = z_1$. That is, for all $z \in \mathbb{C}$, $\text{rank } P^{\mathcal{I}_a}(z_1) \geq \text{rank } P^{\mathcal{I}_a}(z)$. This implies $\text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{rank } P^{\mathcal{I}_a}(z_1)$. Hence, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z_1) < \text{rank } P^{\mathcal{I}_a}(z_1)$. Thus, for $z = z_1$, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z_1) \neq \text{rank } P^{\mathcal{I}_a}(z_1)$, thereby ending the proof. $\square$

We need the following proposition for proving Theorem 4

Proposition 6. Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$. If for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$ the corresponding matrix pencils satisfy $\text{rank } P^{\mathcal{I}_a} = \text{rank } P^{\mathcal{I}_a \setminus i_1} \forall z \in \mathbb{C}$, then, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \leq p$. $\blacksquare$

Proof. Fix a set $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$. Suppose that condition is satisfied for one nonzero realization, say $(W^1, \tilde{B}_a^1, C^1, \tilde{D}_a^1) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$. That is, the corresponding matrix pencil, $P_1^{\mathcal{I}_a}(z)$, satisfies $\text{rank } P_1^{\mathcal{I}_a}(z) = \text{rank } P_1^{\mathcal{I}_a \setminus i_1}(z) \forall z \in \mathbb{C}$. This further implies, from Lemma 5, that $\text{normal-rank } P_1^{\mathcal{I}_a}(z) = \text{normal-rank } P_1^{\mathcal{I}_a \setminus i_1}(z)$. Hence, $\text{normal-rank } G_1^{\mathcal{I}_a}(z) = \text{normal-rank } G_1^{\mathcal{I}_a \setminus i_1}(z)$. Therefore from Milošević et al. (2020, Prop. 1), with respect to the nonzero realization $(W^1, \tilde{B}_a^1, C^1, \tilde{D}_a^1)$, a perfectly undetectable attack using i_1 can be conducted with p elements. Since the condition is satisfied for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, it follows that a perfectly undetectable attack using i_1 can be conducted with p elements for all nonzero realizations. Hence, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \leq p$. $\square$

Notice that, in the sequel, we use the terms “nonzero choices of edge weights of $\mathcal{G}$ ” and “nonzero choice of free parameters in $\mathbb{R}^\alpha$ ” interchangeably.

Proof of Theorem 4. Before proceeding, the following understanding regarding Lemma 4 is crucial: Item (i) ensures that the claim in Lemma 4 holds for z equals zero (see Hershkowitz and Schneider (1993, Thm. 3.9)), whereas item (ii) does the same for all z different from zero.

Let $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$. Note that $\tilde{\mathcal{H}}$ is the bipartite graph associated with the structured system $\begin{bmatrix} W_\omega & \tilde{B}_\beta \\ C_\gamma & \tilde{D}_\sigma \end{bmatrix}$, and that $\tilde{\mathcal{H}}_\times$ is the bipartite graph associated with the structured system $\begin{bmatrix} W_\omega - zI_N & \tilde{B}_\beta \\ C_\gamma & \tilde{D}_\sigma \end{bmatrix}$.

Suppose that, with respect to set $\mathcal{I}_a$, both the conditions in Theorem 4 are satisfied. Then, for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, item (i) in Lemma 4 ensures that $\text{rank } P^{\mathcal{I}_a}(0) = r$, while item (ii) in Lemma 4 ensures that, for $z \neq 0$, $\text{rank } P^{\mathcal{I}_a}(z) = r$.

Since there exists a uniquely restricted matching $\tilde{\mathcal{M}}$ of size r in $\tilde{\mathcal{H}}$ (resp. $\tilde{\mathcal{M}}_\times$ of size r in $\tilde{\mathcal{H}}_\times$ that satisfies $\tilde{\mathcal{M}}_\times \cap \mathcal{E}_{\text{loop}} = \emptyset$), it follows that there exists a uniquely restricted matching $\tilde{\mathcal{M}}$ in $\tilde{\mathcal{H}}_{-i_1}$ with size at least $r - 1$ but no greater than r (resp. $\tilde{\mathcal{M}}_\times$ in $\tilde{\mathcal{H}}_{\times - i_1}$ with size at least $r - 1$, but no greater than r ,

that satisfies $\hat{\mathcal{M}}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$). By assumption, the maximum size of a uniquely restricted matching $\hat{\mathcal{M}}$ in $\tilde{\mathcal{H}}_{-i_1}$ (resp. $\hat{\mathcal{M}}_x$ in $\tilde{\mathcal{H}}_{x-i_1}$ such that $\hat{\mathcal{M}}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$) equals r . Therefore, for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, from items (i) and (iii) in Lemma 4, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(0) = r$, and, for $z \neq 0$, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) = r$.

Thus, for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$,

$$\text{rank } P^{\mathcal{I}_a}(0) = \text{rank } P^{\mathcal{I}_a \setminus i_1}(0) \quad (3)$$

$$\text{rank } P^{\mathcal{I}_a}(z) = \text{rank } P^{\mathcal{I}_a \setminus i_1}(z), \quad \forall z \in \mathbb{C} \setminus 0 \quad (4)$$

From (3) and (4), the following is immediate: for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$,

$$\text{rank } P^{\mathcal{I}_a}(z) = \text{rank } P^{\mathcal{I}_a \setminus i_1}(z), \quad \forall z \in \mathbb{C}$$

Hence, the condition in Proposition 6 is satisfied, which implies that, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \leq p$, thereby ending the proof. $\square$

We need the following lemma for proving Theorem 5.

Lemma 6. *If $\text{rank } P^{\mathcal{I}_a}(z) > \text{rank } P^{\mathcal{I}_a \setminus i_1}(z)$ for all $z \in \mathbb{C}$, then $\text{normal-rank } P^{\mathcal{I}_a}(z) > \text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z)$. ■*

Proof. Suppose that $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{rank } P^{\mathcal{I}_a}(z)$ for all $z \in \mathbb{C}$. Then, for all $z \in \mathbb{C}$, $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) < \max_{z \in \mathbb{C}} \text{rank } P^{\mathcal{I}_a}(z)$, which, from the definition of normal rank of a matrix pencil, implies $\text{rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{normal-rank } P^{\mathcal{I}_a}(z)$. Consequently, $\max_{z \in \mathbb{C}} \text{rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{normal-rank } P^{\mathcal{I}_a}(z)$, and, hence, $\text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z) < \text{normal-rank } P^{\mathcal{I}_a}(z)$. ■

We require the following proposition for proving Theorem 5.

Proposition 7. *Let $p \in \mathbb{Z}_+$ such that $1 \leq p \leq |\mathcal{I}|$. If for every $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$, and for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the corresponding matrix pencils satisfy*

$$\text{rank } P^{\mathcal{I}_a} > \text{rank } P^{\mathcal{I}_a \setminus i_1} \quad \forall z \in \mathbb{C},$$

then, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \geq p + 1$. ■

Proof. Suppose that the condition in Proposition 7 is satisfied for a set $\mathcal{I}_a$. This implies, from Lemma 6, that $\text{normal-rank } P^{\mathcal{I}_a}(z) > \text{normal-rank } P^{\mathcal{I}_a \setminus i_1}(z)$, and, hence, $\text{normal-rank } G^{\mathcal{I}_a}(z) > \text{normal-rank } G^{\mathcal{I}_a \setminus i_1}(z)$. Therefore, from Milošević et al. (2020, Prop. 1) a perfectly undetectable attack, in which component i_1 is used, cannot be conducted using elements in such a set $\mathcal{I}_a$. Since, by assumption, the condition is satisfied for every set $\mathcal{I}_a$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$, it follows that no nonzero realization can be attacked, using i_1 , with exactly p elements. Hence, the minimum number of elements needed for conducting a perfectly undetectable attack using i_1 against any nonzero realization is bounded from below by $p + 1$. Therefore, it follows that, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \geq p + 1$. $\square$

Proof of Theorem 5. The proof technique is quite similar – up to a switching of the inequality, and corresponding reversal of arguments – to that of Theorem 4, and relies on the algebraic conditions given in Proposition 7. The details are skipped in the interest of space.

Proof of Corollary 2. Consider an arbitrary set $\mathcal{I}'_a \subseteq \mathcal{I}$ such that $i_1 \in \mathcal{I}'_a$ and $|\mathcal{I}'_a| = p$. We number this set as q , where

$q \in \{1, 2, \dots, \ell\}$, and note that $\tilde{\mathcal{H}}^q$ and $\tilde{\mathcal{H}}^q_x$ are the associated bipartite graphs.

Proof of item (i): Suppose that there exists a uniquely restricted matching, say $\tilde{\mathcal{M}}^q$ of size $N + p$ in $\tilde{\mathcal{H}}^q$. Then, from item (i) in Lemma 4, it follows that for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the corresponding matrix pencils satisfy $\text{rank } P^{\mathcal{I}'_a}(0) = N + p$. Note that since $|\mathcal{I}'_a| = p$, the number of columns in $P^{\mathcal{I}'_a}$ equals $N + p$. This implies that for every nonzero realization, $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, $\text{normal-rank } P^{\mathcal{I}'_a}(z) = N + p$. Observe that the number of columns in $P^{\mathcal{I}'_a \setminus i_1}$ equals $N + p - 1$, and hence for every nonzero realization $\text{normal-rank } P^{\mathcal{I}'_a \setminus i_1}(z) \leq N + p - 1$, which further implies that $\text{normal-rank } P^{\mathcal{I}'_a}(z) \neq \text{normal-rank } P^{\mathcal{I}'_a \setminus i_1}(z)$. Thus, from Milošević et al. (2020, Prop. 1), a perfectly undetectable attack, using i_1 together with the other elements in set $\mathcal{I}'_a$, cannot be conducted for any nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$.

By assumption, condition in item (i) is satisfied for every $\mathcal{I}_a \subseteq \mathcal{I}$ such that a) $|\mathcal{I}_a| = p$, and b) $i_1 \in \mathcal{I}_a$. Thus, using the same arguments as in for $\mathcal{I}'_a$, it is immediate that a perfectly undetectable attack cannot be conducted using any set of cardinality p that includes i_1 , for any nonzero realization. Therefore, by definition of security index, it follows that, for all nonzero choices of free parameters in $\mathbb{R}^\alpha$, $\delta(i_1) \geq p + 1$.

Proof of item (ii): Suppose that there exists a uniquely restricted matching, say $\tilde{\mathcal{M}}^q_x$ of size $N + p$ in $\tilde{\mathcal{H}}^q_x$. Then, from item (ii) in Lemma 4, for every nonzero realization $(W, \tilde{B}_a, C, \tilde{D}_a) \in (W_\omega, \tilde{B}_\beta, C_\gamma, \tilde{D}_\sigma)$, the corresponding matrix pencils satisfy $\text{rank } P^{\mathcal{I}'_a}(z) = N + p$ for all $z \in \mathbb{C}$ except possibly $z = w_{ii}$ for some (or all) $i \in \{1, 2, \dots, N\}$. The rest of the proof is same as that of item (i), and, therefore, skipped. $\square$

Proof of Proposition 2. Note that, with respect to a given candidate set, all the conditions given in Theorem 5 (resp. Theorem 4) essentially seek negative (resp. positive) answer to the following question: for a given bipartite graph and an appropriately-defined subgraph of the said bipartite graph, for some given set of edges, is the maximum size of a uniquely restricted matching, formed by using only the given edges, the same? Note that answering this question entails finding the maximum size of a uniquely restricted matching, which is known to be NP-hard; see Columbic, Hirst, and Lewenstein (2001, Theorem 3.3). This completes the proof.

References

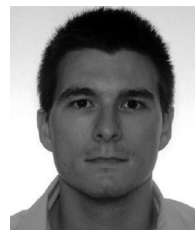
- Ahmed, C. M., Murguia, C., & Ruths, J. (2017). Model-based attack detection scheme for smart water distribution networks. In *Proceedings of the 2017 ACM on Asia Conference on Computer and Communications Security* (pp. 101–113).
- Amin, S. M., & Wollenberg, B. F. (2005). Toward a smart grid: power delivery for the 21st century. *IEEE Power and Energy Magazine*, 3(5), 34–41.
- Andersson, G., Donalek, P., Farmer, R., Hatziaargyriou, N., Kamwa, I., Kundur, P., et al. (2005). Causes of the 2003 major grid blackouts in North America and Europe, and recommended means to improve system dynamic performance. *IEEE Transactions on Power Systems*, 20(4), 1922–1928.
- Bai, C.-Z., Pasqualetti, F., & Gupta, V. (2015). Security in stochastic control systems: Fundamental limitations and performance bounds. In *2015 American Control Conference (ACC)* (pp. 195–200). IEEE.
- Bai, C.-Z., Pasqualetti, F., & Gupta, V. (2017). Data-injection attacks in stochastic control systems: Detectability and performance tradeoffs. *Automatica*, 82, 251–260.
- Chapman, A., & Mesbahi, M. (2013). On strong structural controllability of networked systems: a constrained matching approach. In *Proc. 2013 American Control Conference (ACC)*, Washington, DC, USA (pp. 6126–6131).
- Chong, M. S., Sandberg, H., & Teixeira, A. M. (2019). A tutorial introduction to security and privacy for cyber-physical systems. In *2019 18th European Control Conference (ECC)* (pp. 968–978). IEEE.
- Commault, C., Dion, J.-M., & van der Woude, J. W. (2002). Characterization of generic properties of linear structured systems for efficient computations. *Kybernetika*, 38(5), 503–520.

- Dehghat, M., Ugrinovskii, V., Shames, I., & Langbort, C. (2019). Detection and mitigation of biasing attacks on distributed estimation networks. *Automatica*, 99, 369–381.
- Dion, J.-M., Commault, C., & Van Der Woude, J. (2003). Generic properties and control of linear structured systems: a survey. *Automatica*, 39(7), 1125–1144.
- Fawzi, H., Tabuada, P., & Diggavi, S. (2014). Secure estimation and control for cyber-physical systems under adversarial attacks. *IEEE Transactions on Automatic Control*, 59(6), 1454–1467.
- Giraldo, J., Sarkar, E., Cárdenas, A., Maniatakos, M., & Kantarcioglu, M. (2017). Security and privacy in cyber-physical systems: A survey of surveys. *IEEE Design Test*, 34(4), 7–17.
- Golumbic, M. C., Hirst, T., & Lewenstein, M. (2001). Uniquely restricted matchings. *Algorithmica*, 31(2), 139–154.
- Gracy, S., Garin, F., & Kibangou, A. (2019). Strongly structural input and state observability for linear time invariant network systems. In *European Control Conference, (ECC), Naples, Italy*.
- Gracy, S., Milošević, J., & Sandberg, H. (2021). Security index based on perfectly undetectable attacks: Graph-theoretic conditions- supplementary material. <https://www.diva-portal.org/smash/get/diva2:1584536/FULLTEXT01.pdf>.
- Gracy, S., Milošević, J., & Sandberg, H. (2020). Actuator security index for structured systems. In *2020 American Control Conference (ACC)* (pp. 2993–2998).
- Hartung, C., Reißig, G., & Svaricek, F. (2012). Characterization of strong structural controllability of uncertain linear time-varying discrete-time systems. In *51st IEEE Conference on Decision and Control (CDC), Maui, Hawaii, USA* (pp. 2189–2194).
- Hershkowitz, D., & Schneider, H. (1993). Ranks of zero patterns and sign patterns. *Linear and Multilinear Algebra*, 34(1), 3–19.
- Hopcroft, J. E., & Karp, R. M. (1973). An $n^2/2$ algorithm for maximum matchings in bipartite graphs. *SIAM Journal on Computing*, 2(4), 225–231.
- Kushner, D. (2013). The real story of Stuxnet. *IEEE Spectrum*, 50(3), 48–53.
- Milošević, J., Teixeira, A., Johansson, K. H., & Sandberg, H. (2020). Actuator security indices based on perfect undetectability: Computation, robustness, and sensor placement. *IEEE Transactions on Automatic Control*, 65(9), 3816–3831.
- Mo, Y., Garone, E., Casavola, A., & Sinopoli, B. (2010). False data injection attacks against state estimation in wireless sensor networks. In *49th IEEE Conference on Decision and Control (CDC)* (pp. 5967–5972). IEEE.
- Mo, Y., Kim, T. H. J., Brancik, K., Dickinson, D., Lee, H., Perrig, A., et al. (2012). Cyber-physical security of a smart grid infrastructure. *Proceedings of the IEEE*, 100(1), 195–209.
- Mohsenian-Rad, A.-H., & Leon-Garcia, A. (2011). Distributed internet-based load altering attacks against smart power grids. *IEEE Transactions on Smart Grid*, 2(4), 667–674.
- Murguia, C., Shames, I., Ruths, J., & Nesic, D. (2018). Security metrics of networked control systems under sensor attacks (extended preprint). arXiv preprint arXiv:1809.01808.
- Olfati-Saber, R., & Murray, R. (2002). Distributed cooperative control of multiple vehicle formations using structural potential functions. *IFAC Proceedings Volumes*, 35(1), 495–500.
- Pasqualetti, F., Bicchi, A., & Bullo, F. (2012). Consensus computation in unreliable networks: A system theoretic approach. *IEEE Transactions on Automatic Control*, 57(1), 90–104.
- Penso, L., Rautenbach, D., & dos Santos Souza, U. (2015). Graphs in which some and every maximum matching is uniquely restricted. *Journal of Graph Theory*.
- Sandberg, H., & Teixeira, A. M. (2016). From control system security indices to attack identifiability. In *2016 Science of Security for Cyber-Physical Systems Workshop (SOSCYPS)* (pp. 1–6). IEEE.
- Sinopoli, B., Sharp, C., Schenato, L., Schaffert, S., & Sastry, S. (2003). Distributed control applications within sensor networks. *Proceedings of the IEEE*, 91(8), 1235–1246.
- Slay, J., & Miller, M. (2007). Lessons learned from the Maroochy water breach. In *Proceedings of the International Conference on Critical Infrastructure Protection*.
- Sou, K., Sandberg, H., & Johansson, K. H. (2012). Computing critical k -tuples in power networks. *IEEE Transactions on Power Systems*, 27(3), 1511–1520.

- Sundaram, S. (2018). Fault-tolerant and secure control systems. In *Lecture notes*, University of Waterloo.
- Sundaram, S., & Hadjicostis, C. N. (2010). Distributed function calculation via linear iterative strategies in the presence of malicious agents. *IEEE Transactions on Automatic Control*, 56(7), 1495–1508.
- Tang, Z., Kuijper, M., Chong, M. S., Mareels, I., & Leckie, C. (2019). Linear system security—Detection and correction of adversarial sensor attacks in the noise-free case. *Automatica*, 101, 53–59.
- Vukovic, O., Sou, K., Dan, G., & Sandberg, H. (2012). Network-aware mitigation of data integrity attacks on power system state estimation. *IEEE Journal on Selected Areas in Communications*, 30(6), 1108–1118.
- Weerakkody, S., Liu, X., Son, S., & Sinopoli, B. (2016). A graph-theoretic characterization of perfect attackability for secure design of distributed control systems. *IEEE Transactions on Control of Network Systems*, 4(1), 60–70.
- van der Woude, J. (1999). The generic number of invariant zeros of a structured linear system. *SIAM Journal on Control and Optimization*, 38(1), 1–21.



Sebin Gracy is a Post-Doctoral Researcher in the Division of Decision and Control Systems in the School of Electrical Engineering and Computer Science at KTH Royal Institute of Technology. He obtained his Ph.D. degree at Université Grenoble-Alpes in November, 2018. Prior to that, he obtained his M.S. and B.E. degrees in Electrical Engineering from the University of Colorado at Boulder and the University of Mumbai, in December, 2013 and June 2010, respectively. His research interests are in the realm of networked control systems.



Jezdimir Milošević received the M.Sc. degree in electrical engineering and computer science in 2015 from the School of Electrical Engineering, University of Belgrade, Belgrade, Serbia, and the Ph.D. degree in automatic control from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2020. He was a Visiting Researcher with the University of Hawaii at Manoa in 2014, and Massachusetts Institute of Technology in 2018 and 2019. He is currently a research engineer at Scania Autonomous Transport Solutions, Södertälje, Sweden. His research interests are within planning and decision making algorithms for autonomous heavy-duty vehicles.



Henrik Sandberg is Professor at the Division of Decision and Control Systems, KTH Royal Institute of Technology, Stockholm, Sweden. He received the M.Sc. degree in engineering physics and the Ph.D. degree in automatic control from Lund University, Lund, Sweden, in 1999 and 2004, respectively. From 2005 to 2007, he was a Post-Doctoral Scholar at the California Institute of Technology, Pasadena, USA. In 2013, he was a visiting scholar at the Laboratory for Information and Decision Systems (LIDS) at MIT, Cambridge, USA. He has also held visiting appointments at the Australian National University and the University of Melbourne, Australia. His current research interests include security of cyber-physical systems, power systems, model reduction, and fundamental limitations in control. Dr. Sandberg was a recipient of the Best Student Paper Award from the IEEE Conference on Decision and Control in 2004, an Ingvar Carlsson Award from the Swedish Foundation for Strategic Research in 2007, and a Consolidator Grant from the Swedish Research Council in 2016. He has served on the editorial boards of IEEE Transactions on Automatic Control and the IFAC Journal Automatica.